

Options Pricing Mini-Project Report

This report shows pricing for European, American, and Asian options using various models in Python. Parameters used:

$S_0=100$, $K=100$, $r=0.05$, $\sigma=0.2$, $T=1.0$

Option Prices:

Black-Scholes Call: 10.4506

Black-Scholes Put : 5.5735

American Put (Binomial): 6.0864

Asian Call (Monte Carlo): 5.7507

European Call (Monte Carlo): 10.5070

Greeks (Call, Black-Scholes):

Delta=0.6368, Gamma=0.0188, Vega=37.5240, Theta=-6.4140, Rho=53.2325

Plots:

